ECON 1550 Problem Set 8 Answer Key

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Submission:	Canvas or Gradescope

1. **Chapter 6: Output and the Exchange Rate in the Long Run**

Question 1 from Chapter 4 of the textbook is:

> "Suppose there is a reduction in aggregate real money demand, that is, a negative shift in the aggregate real money demand function. Trace the short- and long-run effects on the exchange rate, interest rate, and price level."

In Problem Set 4, question 1, part (a), you were asked to answer that question, but only for the short run, and treating the expected exchange rate and the price level as exogenous.

Since you were asked to only focus on the short run and to treat the expected exchange rate as exogenous, knowing whether the shift in aggregate real money demand was temporary or permanent was irrelevant (the answer is the same in either case).

Now that we have developed the AA-DD model, we are better equipped to tackle the long-run effects part of the question and understand the difference between temporary and permanent changes in exogenous variables.

Answer the questions below using the AA-DD model given by the following equations:

$$\text{DD Schedule:} \quad Y = \mathrm{C}(\underset{(+)}{Y - T}) + I + G + \mathrm{CA}(\underset{(+)}{EP^*/P}, \underset{(-)}{Y - T}, \underset{(+)}{Y^*})$$

$$\text{AA Schedule:} \quad \frac{M^s}{P} = \mathrm{L}(\underset{(-)}{R^* + E^e/E - 1}, \underset{(+)}{Y})$$

where the exogenous variables are in the tables below.

AA-DD model: exogenous variables

Variable	Description
T	taxes
I	investment
G	government spending
M^s	money supply
Y^f	full-employment level of output
Y^*	foreign income
P^*	foreign price level
R^*	foreign nominal interest rate

AA-DD model: endogenous variables

Variable	Description
Y	income, production
C	consumption
CA	current account
E	nominal exchange rate
E^e	expected nominal exchange rate
P	domestic price level

As usual, we assume that:

- in the long run, $Y = Y^f$, $E = E^e$, and $P = P^e$;
- in the short run, P is fixed, and E^e is equal to the value of E in the final long-run equilibrium.

Starting from an initial long-run equilibrium, there is an unanticipated reduction in the real money demand function from $\mathrm{L}(\cdot,\cdot)$ to $\mathrm{L}(\cdot,\cdot) - 1$.

(a) Assuming the change in the money demand function is temporary, sketch in an AA-DD diagram the initial equilibrium, the short-run equilibrium, the long-run equilibrium, and the transition from the short run back to the long run. A qualitatively correct sketch is enough. Explain in words how and why the curves shift, or do not shift, at each point in time.

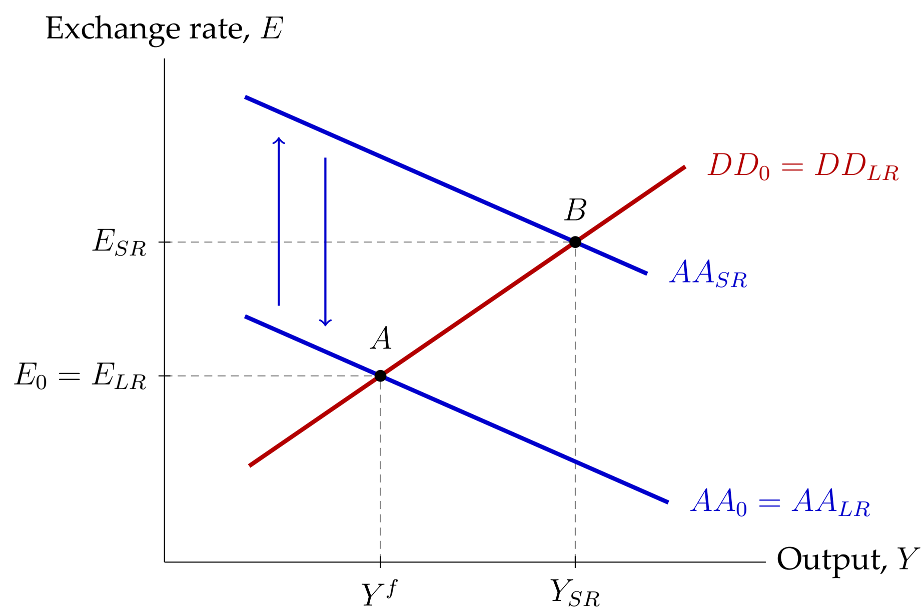
Solution: Because the change in money demand is temporary, the final long-run equilibrium is the same as the initial one. Hence E^e does not change: in the short run, people still expect the exchange rate to return to its initial long-run value E_0.
In the AA-DD diagram:

- Start at point A, the intersection of AA_0 and DD_0, with output Y^f and exchange rate E_0.
- On impact, lower real money demand creates an excess supply of real balances at the initial interest rate and output. To restore money-market equilibrium, the domestic interest rate must fall. By UIP, the lower interest rate implies a depreciation of the domestic currency, so the AA curve shifts up to AA_{SR}.
- The DD curve does not shift. The shock does not directly affect T, I, G, Y^*, P^*, or the current price level P.
- The new short-run equilibrium is point B, where AA_{SR} intersects DD_0. Relative to A, we have $E_{SR} > E_0$ and $Y_{SR} > Y^f$.
- Because the shock is temporary, when money demand returns to its original level the AA curve shifts back down to AA_0. The economy moves back along DD_0 from B to A.

Therefore the long-run equilibrium is the initial equilibrium again:

$$E_{LR} = E_0 \qquad \text{and} \qquad Y_{LR} = Y^f.$$

A correct sketch has one temporary upward shift of AA, then an immediate shift back down once the temporary shock disappears, no shift in DD, and no permanent effect. In the figure, point A is the initial and long-run equilibrium, at the intersection of $AA_0 = AA_{LR}$ and $DD_0 = DD_{LR}$. Point B is the short-run equilibrium, at the intersection of AA_{SR} and DD_0. The blue upward arrow indicates the impact effect: AA shifts up from AA_0 to AA_{SR} when money demand falls. The blue downward arrow indicates the reversal: AA shifts back down from AA_{SR} to AA_0 when the temporary shock ends. The red DD schedule stays fixed throughout. The dashed lines show that output rises from Y^f to Y_{SR} and the exchange rate rises from E_0 to E_{SR}.



(b) Plot the time paths of E, Y, and CA for the temporary change the money demand function considered in (a). A qualitatively correct path is enough.

Solution: Because the shock is temporary, all three variables return to their initial levels in the long run.

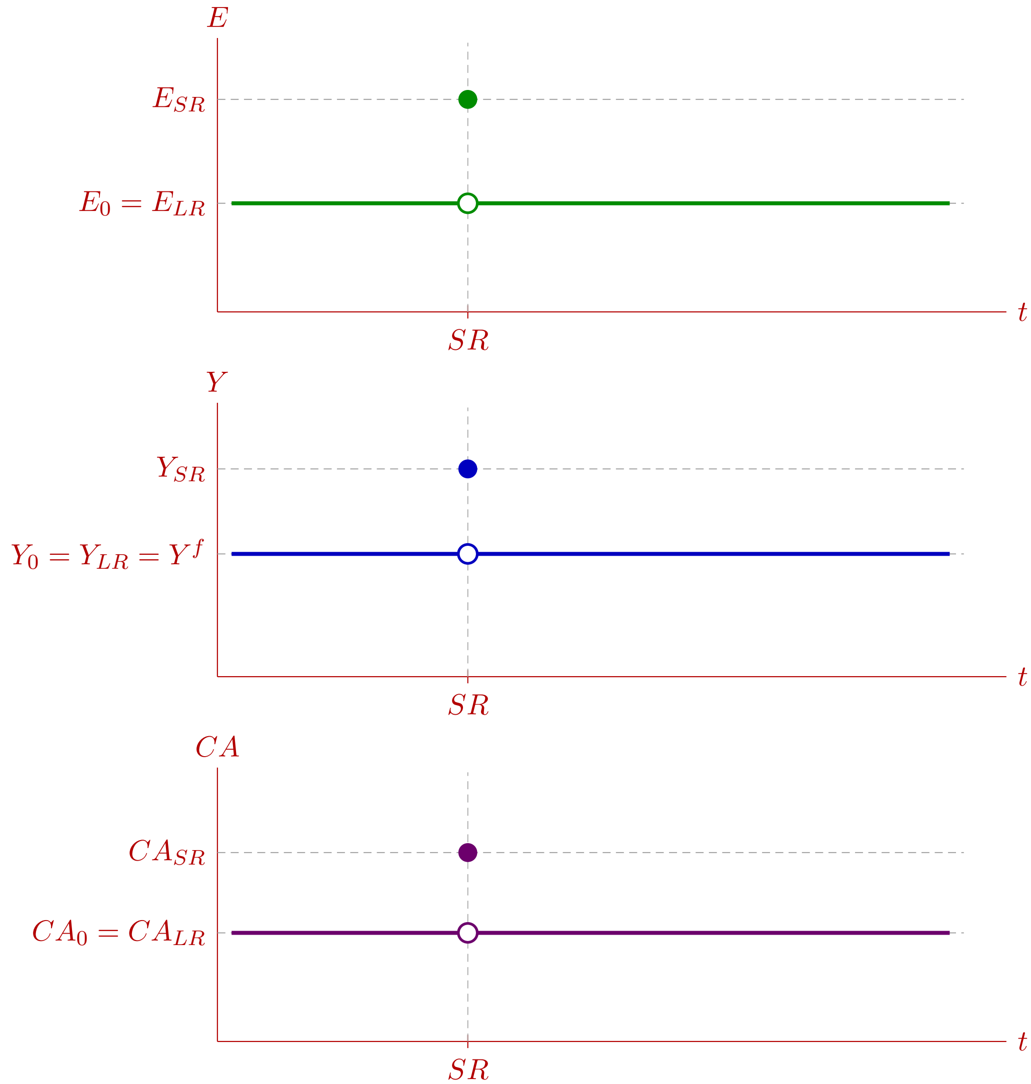
- E: is at E_0 before the temporary shock, jumps up to E_{SR} at the short-run instant, and then jumps back immediately to E_0 once the shock is gone.
- Y: is at Y^f before the shock, jumps up to Y_{SR} at the short-run instant, and then jumps back immediately to Y^f.
- CA: is at its initial level before the shock, jumps up to CA_{SR} at the short-run instant, and then jumps back immediately to its initial level.

To see why CA rises in the short run, use goods-market equilibrium:

$$CA = Y - C(Y - T) - I - G.$$

Since T, I, and G do not change, and C rises by less than one-for-one with Y, a higher short-run level of output implies a higher current account.
Because the temporary money-demand shock is treated as a short-run impact only, the figure shows a single displaced point at SR, not a gradual transition.

In each panel, the dashed vertical line marks that one short-run instant. The open circle shows the value just before the temporary shock hits, and the filled circle shows the short-run value while the shock is present. Immediately after that instant, the variable is back at its initial long-run level, so the path to the right of SR is flat at the original value.



(c) Answer the same question as in part (a), but assuming the change in the money demand function is permanent.

Solution: Now the reduction in money demand is permanent, so the final long-run equilibrium is different from the initial one.

In the new long run, we still have $Y = Y^f$, $E = E^e$, and $R = R^*$. The money market must therefore satisfy

$$\frac{M^s}{P_{LR}} = L(R^*, Y^f) - 1,$$

whereas initially

$$\frac{M^s}{P_0} = L(R^*, Y^f).$$

Since money demand is now permanently lower, the right-hand side is smaller, so $P_{LR} > P_0$: the price level must rise to reduce real balances.

In the goods market, long-run output is still Y^f, and T, I, G, Y^*, and P^* are unchanged. Therefore the current account in the long run must be the same as initially:

$$CA_{LR} = Y^f - C(Y^f - T) - I - G = CA_0.$$

Since CA depends positively on the real exchange rate $q = EP^*/P$, this implies $q_{LR} = q_0$. Because $P_{LR} > P_0$, the nominal exchange rate must also be permanently higher:

$$E_{LR} > E_0.$$

So expected future exchange rate E^e rises immediately to this new long-run level.

In the AA-DD diagram:

- Start at point A, the intersection of AA_0 and DD_0, with output Y^f and exchange rate E_0.
- On impact, AA shifts up for two reasons. First, lower money demand lowers the interest rate required for money-market equilibrium. Second, the higher expected long-run exchange rate $E^e = E_{LR}$ also raises today's exchange rate through UIP.
- DD does not shift on impact because the price level is fixed in the short run.
- The new short-run equilibrium is point B, above and to the right of A. Thus $E_{SR} > E_0$ and $Y_{SR} > Y^f$. In fact,

$$E_{SR} > E_{LR} > E_0,$$

so the exchange rate overshoots its new long-run level.

- Because $Y_{SR} > Y^f$, the price level starts to rise over time. As P rises, the AA curve shifts down because real money balances M^s/P fall. At the same time, the DD curve shifts left because, for any given nominal exchange rate E, a higher P lowers the real exchange rate $q = EP^*/P$ and reduces demand for home output.

- The economy moves from B back to full employment at point C, where the new long-run curves intersect. At C, output is again Y^f, the price level is higher, and the nominal exchange rate is higher than initially but below its short-run peak.

Figure 1 shows the impact effect. Point A is the initial equilibrium, where AA_0 intersects DD_0, with output Y^f and exchange rate E_0. On impact, the AA curve jumps up from AA_0 to AA_{SR} (the single-headed blue upward arrow). The DD curve does not shift. Point B is the short-run equilibrium, at the intersection of AA_{SR} and DD_0, with output Y_{SR} and exchange rate E_{SR}.
Figure 2 shows the transition from the short run to the long run. The economy is always in equilibrium, always at the intersection of the AA and DD curves. Between the short run and the long run, both curves shift continuously as the price level rises. The AA curve shifts gradually downward from AA_{SR} to AA_{LR} (the multi-headed blue downward arrows) because rising prices reduce real money balances M^s/P. The DD curve shifts gradually to the left from DD_0 to DD_{LR} (the multi-headed red leftward arrows) because, for any given E, a higher P lowers the real exchange rate $q = EP^*/P$ and reduces demand for home output. The black curved path from B to C traces the equilibrium trajectory: it shows how the intersection of the AA and DD curves moves as both curves shift simultaneously over time. At each instant along this path, the economy sits at the intersection of the current AA and DD curves. Point C is the new long-run equilibrium, where AA_{LR} intersects DD_{LR}, with output back at Y^f and the exchange rate at E_{LR}, which is above E_0 but below the short-run peak E_{SR}.

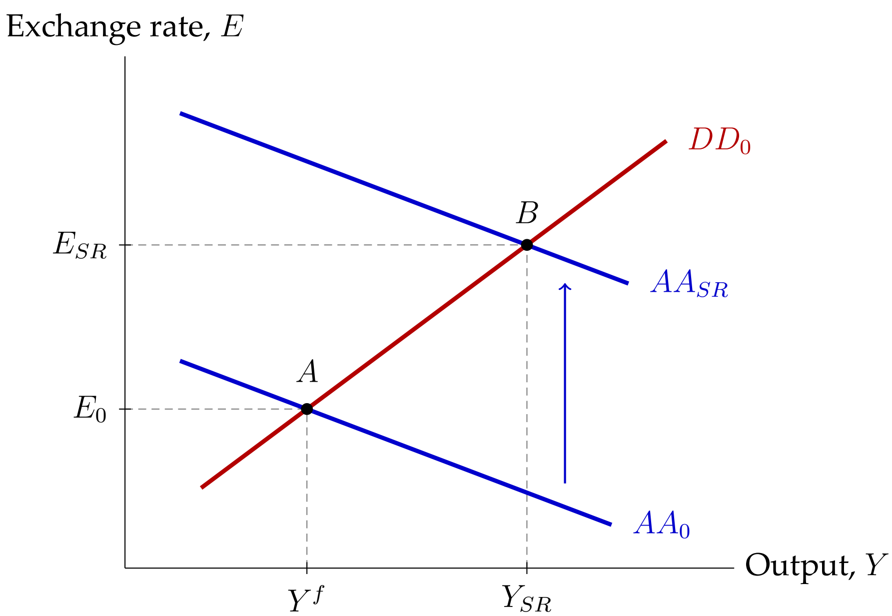


Figure 1: Permanent reduction in money demand: impact effect (initial equilibrium to short run).

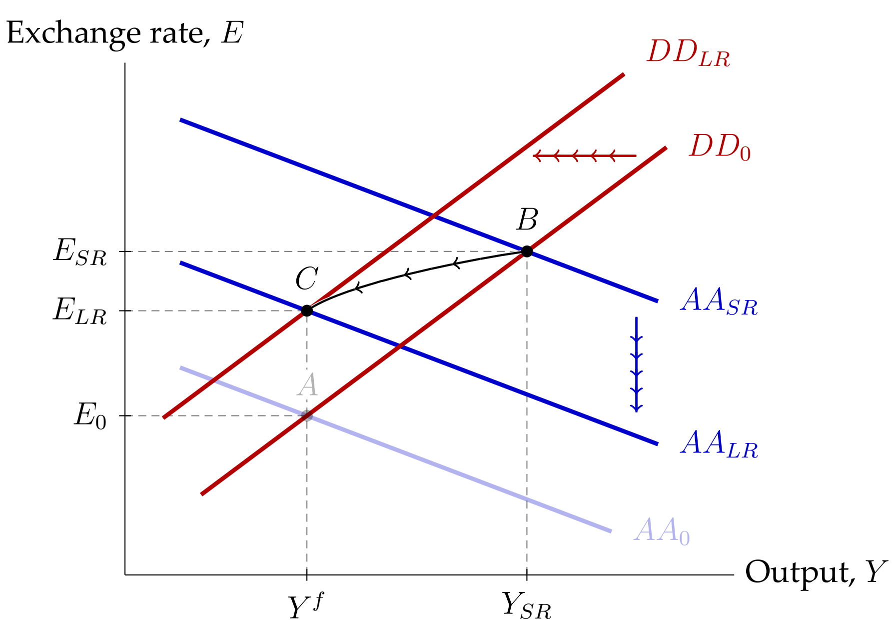


Figure 2: Permanent reduction in money demand: transition from the short run to the long run.

(d) Answer the same question as in part (b), but assuming the change in the money demand function is permanent.

Solution: The time paths are:

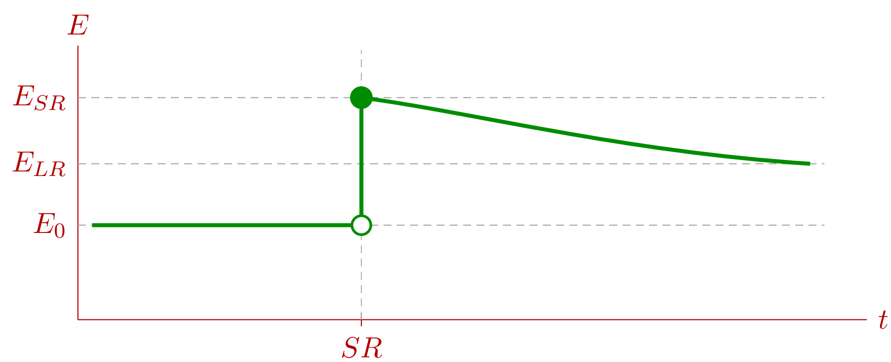
- E: jumps up on impact to its most depreciated level, then gradually falls back to a new long-run level that is still above the initial one. Thus

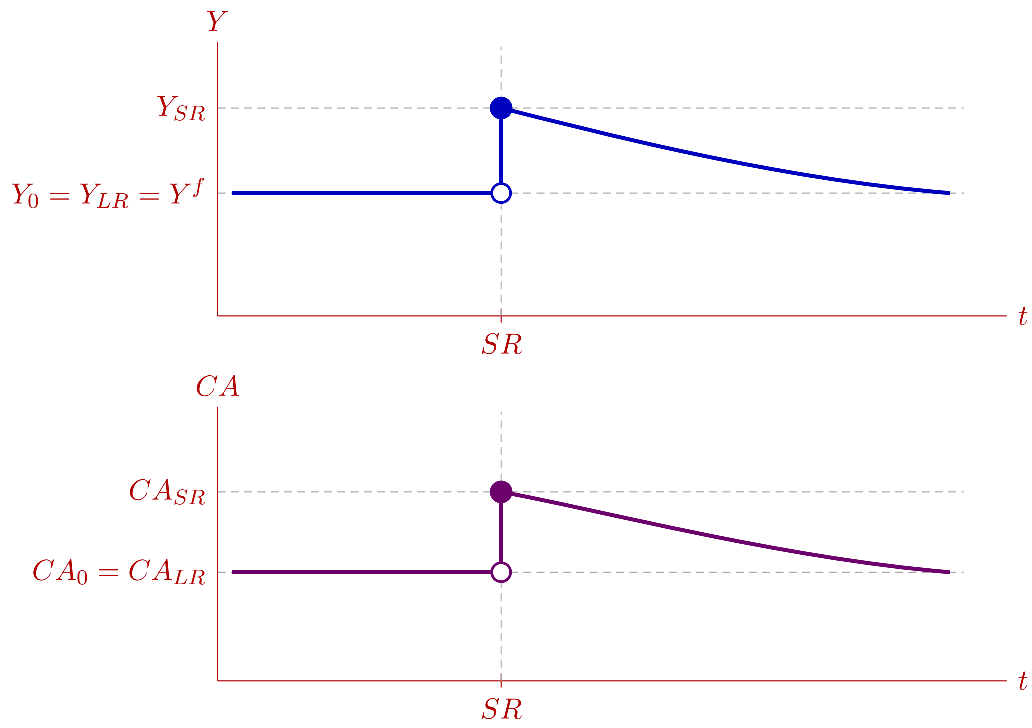
$$E_{SR} > E_{LR} > E_0.$$

- Y: jumps above Y^f in the short run, then gradually declines back to Y^f as prices rise.
- CA: rises on impact, then gradually falls back to its initial level.

The long-run current account equals its initial level because in the long run we are again at $Y = Y^f$, while T, I, G, and Y^* are unchanged. Therefore the goods market requires the same CA as before.

So, relative to the temporary case, the difference is that E now has a permanent increase, and the initial depreciation is larger because the exchange rate overshoots its new long-run level. In each panel, the dashed vertical line marks the short-run instant. The open circle on that line shows the value just before the shock, and the filled circle shows the immediate short-run value after the shock. To the right of that instant, the economy transitions toward the long run. For E, the path converges to a level above E_0; for Y and CA, the paths converge back to their initial long-run levels.





(e) Consider the same permanent reduction in real money demand as in parts (c) and (d), but now impose the Problem Set 4 assumption that both the expected exchange rate and the price level are exogenous. Explain what "goes wrong" under these assumptions.

Hint 1: We do not define what exactly we mean by "goes wrong" intentionally, as there are many ways in which this exercise is problematic. As you work through this question, you should run into trouble in some way. For example, you may find the model contradicts itself, the outcome is non-sensical, the intuition of behavioral equations is not respected.

Hint 2: This question is intended to make you think about why it was not a good idea to tackle the long run in Problem Set 4.

Solution: What goes wrong is simple: once both P and E^e are exogenous, the model generally has no long-run equilibrium after a permanent fall in money demand.

After the shock, at $Y = Y^f$ and $R = R^*$, people want to hold less real money than before. But if P is fixed from outside, real money supply M^s/P is also fixed from outside. So the money market will usually not clear at the old long-run conditions.

Could the exchange rate adjust instead? If E rises permanently while P is fixed, then the real exchange rate

$$q = \frac{EP^*}{P}$$

also rises permanently. That raises net exports and keeps output above Y^f. So we do not get a long-run equilibrium with full employment.

Could we rescue the exercise by choosing an exogenous path for P and an exogenous path for E^e? That does not help either. A model solution should work for any given values, or paths, of the exogenous variables. Here, equilibrium would exist only if we hand-pick a very special path for P and E^e after seeing the shock. If we changed that path a little bit, equilibrium would disappear. That is not the model solving the problem; that is us imposing the answer from outside.

This is why it was not a good idea to tackle the long run in Problem Set 4. There, fixing P and E^e was fine because the question was only about the short run. For the long run, P and E^e are exactly the variables that need to adjust.

(f) In the scenario from part (e), keep the expected exchange rate exogenous, but allow the price level to be endogenous. Does this fix the problem? Explain.

Solution: No. Allowing P to be endogenous fixes only one part of the problem. Now P can rise, so real balances M^s/P can fall and the money market can clear. But E^e is still exogenous, and that is still a problem.

After a permanent fall in money demand, the long-run real exchange rate must return to its initial value, while the long-run price level is higher. That means the long-run nominal exchange rate must also be higher:

$$E_{LR} > E_0.$$

So the expected exchange rate must adjust too.

If the exogenous path for E^e does not give exactly that new long-run value, then equilibrium still fails. Either $E = E^e$ in the long run, in which case the real exchange rate is wrong and $Y \neq Y^f$, or $E \neq E^e$, in which case the long-run UIP condition fails.

Could we choose a special exogenous path for E^e to make the model work? Again, that does not really solve the problem. A solution should work for any given exogenous path. If equilibrium exists only for one carefully chosen path of E^e, and disappears when that path changes a little, then the model is not delivering the answer. We are building the answer into the exogenous variables. So this still does not fix the Problem Set 4 issue. That exercise worked only as a short-run exercise. For the long run, E^e cannot be fixed from outside.

2. **Fixing the Real Exchange Rate**

Problem Set 7, question 1, part (a), gives you a preview of how governments can keep the nominal exchange rate fixed at a desired level. Rather than increasing the money supply to fully finance a budget deficit, the central bank can decide to pick the size of the response of the money supply so that the nominal exchange rate remains unchanged. More generally, when the DD curve shifts, the central bank can use monetary policy to shift the AA just the right amount so that the shifted AA and DD intersect at a point that has the same nominal exchange rate as the original AA and DD before either of them shifts.

In this problem, we explore why, unlike the nominal exchange rate, the government can't keep the real exchange rate q fixed at a desired level.

The money market and foreign exchange market equilibrium conditions are

$$\frac{M^s}{P} = L(R, Y) = -5R + 0.8Y, \tag{1}$$

$$R = R^* + \frac{E^e}{E} - 1, \tag{2}$$

where M^s is the exogenous money supply, P is the endogenous domestic price level, Y is endogenous real income, R is the endogenous domestic interest rate, R^* is the exogenous foreign interest rate, E^e is the endogenous expected nominal exchange rate, and E is the endogenous nominal exchange rate.

The DD curve is given by:

$$Y = 0.5(EP^*/P) + 0.75Y. \tag{3}$$

The economy is initially at its long-run equilibrium with output equal to the exogenously given full-employment level of output $Y^f = 4$, and money supply $M_0^s = 2.6$. Assume $P^* = 1$ and $R^* = 0.12 = 12\%$ at all times.

(a) Find the domestic interest rate R_0 in the initial long-run equilibrium.

Solution: From the UIP condition in equation (2), and the long-run equilibrium condition $E^e = E$, we get $R_0 = R^* = 0.12 = 12\%$.

(b) Find the domestic price level P_0 in the initial long-run equilibrium. In the money market equation, if the interest rate is, for example, 5%, use $R = 0.05$ and not 5.

Solution: From the money market equilibrium condition in equation (1), the long-run equilibrium condition $Y_0 = Y^f$, and $R_0 = R^*$ found in part (a),

$$\frac{M_0^s}{P_0} = L(R_0, Y_0) = L(R^*, Y^f) = -5R^* + 0.8Y^f.$$

Solving for P_0, and using $M_0^s = 2.6$, $Y^f = 4$ and $R^* = 0.12$,

$$P_0 = \frac{M_0^s}{-5R^* + 0.8Y^f} = \frac{2.6}{-5 \times 0.12 + 0.8 \times 4} = 1.$$

(c) Find the nominal exchange rate E_0 and the expected nominal exchange rate E_0^e, in the initial long-run equilibrium.

Solution: The DD curve in equation (3) in the initial long-run equilibrium is

$$Y_0 = 0.5E_0P^*/P_0 + 0.75Y_0.$$

Using the long-run equilibrium condition $Y_0 = Y^f$ and solving for E_0 gives

$$0.25Y_0 = 0.5E_0P^*/P_0 \qquad \Rightarrow \qquad E_0 = 0.5Y_0P_0/P^* = 0.5Y^fP_0/P^*.$$

Plugging in the values $Y^f = 4$ and $P^* = 1$ given by the problem, and the value of $P_0 = 1$ found in part (b),

$$E_0 = 0.5 \times 4 \times 1/1 = 2.$$

Since in any long-run equilibrium we have $E = E^e$, the expected nominal exchange rate in the long-run equilibrium is $E_0^e = E_0 = 2$.

(d) Show that the real exchange rate in the initial long-run equilibrium is $q_0 = 2$.

Solution: Using the value $P^* = 1$ given by the problem, the value $P_0 = 1$ found in part (b), the value $E_0 = 2$ found in part (c), and the definition $q = EP^*/P$ of the real exchange rate, we get

$$q_0 = E_0 P^*/P_0 = 2 \times 1/1 = 2. \tag{4}$$

(e) The central bank decides it wants to fix the real exchange rate at a value of $\bar{q} = 1$ forever. To do so, it changes the money supply unexpectedly from its current level of $M_0^s = 2.6$ to some value M_{SR}^s in the short run and some value M_{LR}^s in the long run. Show that there are no values for M_{SR}^s and M_{LR}^s that can implement equilibria with the fixed real exchange rate $q = \bar{q} = 1$.

Hint: Focus on the goods market in the long run.

Solution: In any long-run equilibrium we must have $Y = Y^f = 4$. So apply the DD equation (3) in the long run:

$$4 = 0.5q_{LR} + 0.75 \times 4.$$

Therefore

$$4 = 0.5q_{LR} + 3 \qquad \Rightarrow \qquad q_{LR} = 2.$$

So the goods market says that any long-run equilibrium must have real exchange rate $q_{LR} = 2$. But the policy target is $\bar{q} = 1$. Those two conditions are incompatible.

Equivalently, because $q = EP^*/P$ and $P^* = 1$, a target $q = 1$ requires

$$E_{LR} = P_{LR},$$

while long-run goods-market equilibrium requires

$$\frac{E_{LR}}{P_{LR}} = 2 \qquad \text{or} \qquad E_{LR} = 2P_{LR}.$$

So there is no long-run equilibrium with $q = 1$, no matter what values the central bank chooses for M_{SR}^s and M_{LR}^s. Since the long run already fails, no pair (M_{SR}^s, M_{LR}^s) can implement that policy forever.

(f) You are the finance secretary for a very stubborn president who believes that $q = 1$ can be sustained forever. "This is easy", the president says. "Just increase the money supply as soon as output starts to rise back toward Y^f". Explain to the president why this is a really bad idea.

Solution: There are probably many ways to explain this. Here's one.
If the government tries to keep $q = 1$, the DD equation implies

$$Y = 0.5 \times 1 + 0.75Y,$$

so

$$Y = 2.$$

That is far below full employment, since $Y^f = 4$. So the economy has persistent slack and downward pressure on prices.
As P falls, keeping $q = 1$ requires the nominal exchange rate to fall one-for-one with it, because $q = EP^*/P$ and $P^* = 1$, so

$$q = 1 \qquad \Rightarrow \qquad E = P.$$

Now look at the money market. If there were a long-run equilibrium with $Y = 2$ and $R = R^* = 0.12$, money demand would be

$$\frac{M^s}{P} = -5(0.12) + 0.8(2) = 1.$$

If P keeps falling, money-market equilibrium requires M^s to fall with P, not rise.
That is why the president's idea is backwards. Increasing the money supply pushes the nominal exchange rate up, which pushes the real exchange rate above 1 and moves the economy away from the target. The government would end up chasing a target that is not compatible with long-run equilibrium in the first place.